

# COMP 3311: Database Management Systems

## Movie Streaming Management System

### Entity-Relationship Schema Design Assignment: Questions and Answers

#### Some Design Advice and Guidelines

If this is your first database design it may be difficult to decide what and what not to represent in your E-R schema. A common mistake made by those new to database design is to represent operations or authorization for operations on the data in the database design. A database design normally **does not** represent operations that users can perform on the data (though there are special exceptions). Moreover, authorization of operations is a separate issue from what data to represent. Authorization issues are dealt with by the database administrator who specifies what operations (read, update, insert, delete) each user can perform on each entity/relationship. A database design should represent only the (static) data required by an application that needs to be stored permanently to support any operations performed on the database.

#### Null Values

The need to use null values for attributes should be avoided if at all possible as they waste storage space and affect the result of some operations.

#### Surrogate Keys

Surrogate keys should not be used when designing an E-R schema as they have no meaning in the application domain. Moreover, surrogate keys relate to implementation issues and implementation issues should not be a consideration when designing an E-R schema.